



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Introduction to Cryptography and Security

Public key encryption from Diffie-Hellman

- 1 Diffie–Hellman Key Exchange
- 2 Some Abstract Algebra
- 3 Discrete Logarithm and Diffie-Hellman Problems
- 4 The ElGamal Public-key System

Diffie–Hellman Key Exchange (DHKE)

- For a prime p , exponentiation in $\mathbb{Z}_p^*$ is a one-way function:
 - Easy: $x \mapsto g^x \pmod{p}$
 - Hard: $g^x \pmod{p} \mapsto x$
- Exponentiation “commutes” in the sense that

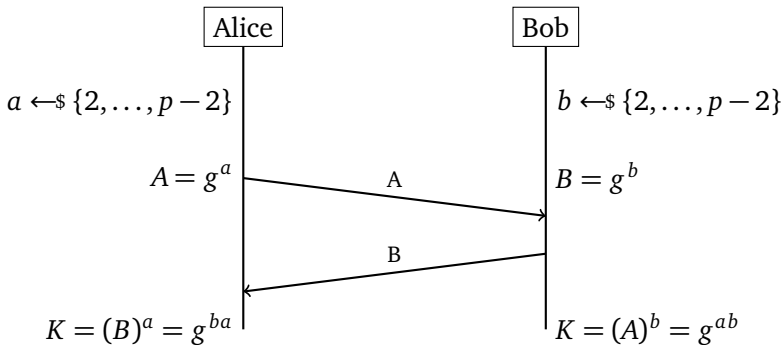
$$k \equiv (g^x)^y \equiv (g^y)^x \equiv g^{xy} \pmod{p}.$$

- The value k is the joint secret in DHKE and can be used as the session key.

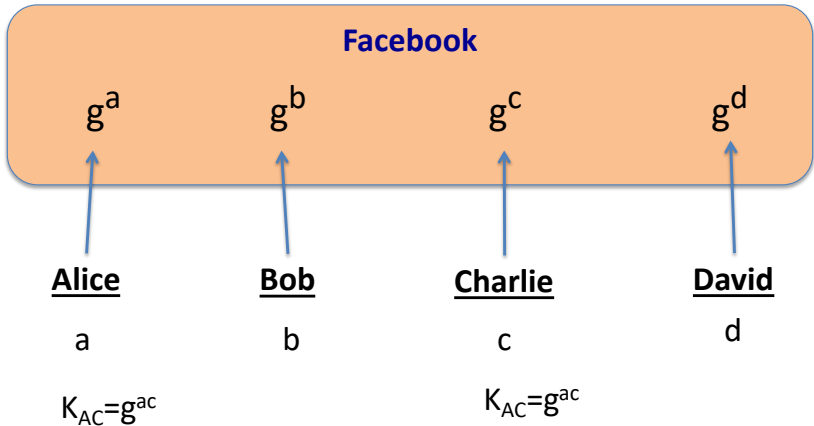
- 1 Choose a large prime p .
- 2 Choose an integer $g \in \{2, 3, \dots, p - 2\}$.
- 3 Publish p and g .

The Diffie-Hellman protocol

Public parameter: g, p



Another look at Diffie-Hellman Protocol



Compute the two public keys and the shared key K for Diffie-Hellman protocol with the parameters $p = 467$ and $g = 2$, and

- ① $a = 3, b = 5$
- ② $a = 400, b = 134$
- ③ $a = 228, b = 57$

Exercise: DH protocol over Galois field $GF(2^m)$

- All arithmetic is done in $GF(2^5)$ with $P(x) = x^5 + x^2 + 1$ as an irreducible field polynomial.
- The generator for Diffie-Hellman protocol is $g = x^2$. The private key are $a = 3$ and $b = 12$.
- What is the shared key K ?

Solution: DH protocol over Galois field $GF(2^m)$

- All arithmetic is done in $GF(2^5)$ with $P(x) = x^5 + x^2 + 1$ as an irreducible field polynomial.
- The generator for Diffie-Hellman protocol is $g = x^2$. The private key are $a = 3$ and $b = 12$.
- What is the shared key K ?

```
1 sage: K.<x>=GF(2^5, name='x', modulus=x^5+x^2+1)
2 sage: K=((x^2)^3)^12
3 sage: K
4 x^4 + 1
```

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Definition

A **group** is a set G with an operation $\circ$ satisfying:

- 1 **Closure**: $a, b \in G \Rightarrow a \circ b \in G$.
- 2 **Associativity**: $a \circ (b \circ c) = (a \circ b) \circ c$.
- 3 **Identity**: There exists $1 \in G$ such that $a \circ 1 = a$.
- 4 **Inverse**: For each $a \in G$, there exists $a^{-1} \in G$ with $a \circ a^{-1} = 1$.

A group G is **abelian** (or commutative) if, furthermore,
 $a \circ b = b \circ a$ for all $a, b \in G$.

A group G is finite if it has a finite number of elements. We denote the cardinality or **order** of the group G by $|G|$.

Theorem

The set $\mathbb{Z}_n^*$, which consists of all integers $i = 1, \dots, n-1$ for which $\gcd(i, n) = 1$, forms an abelian group under multiplication modulo n . The identity element is $e = 1$.

Example

For $n = 9$, $\mathbb{Z}_9^* = \{1, 2, 4, 5, 7, 8\}$. The multiplication table for $\mathbb{Z}_9^*$ is as follows:

$\times \bmod 9$	1	2	4	5	7	8
1	1	2	4	5	7	8
2	2	4	8	1	5	7
4	4	8	7	2	1	5
5	5	1	2	7	8	4
7	7	5	1	8	4	2
8	8	7	5	4	2	1

Conditions for a group are satisfied: closure, identity, inverses, and commutativity. Associativity follows from multiplication in $\mathbb{Z}_n$.

Definition

Consider a group G and $S \subseteq G$. Then S is called a **subgroup** of G if S is a group under the operation of G .

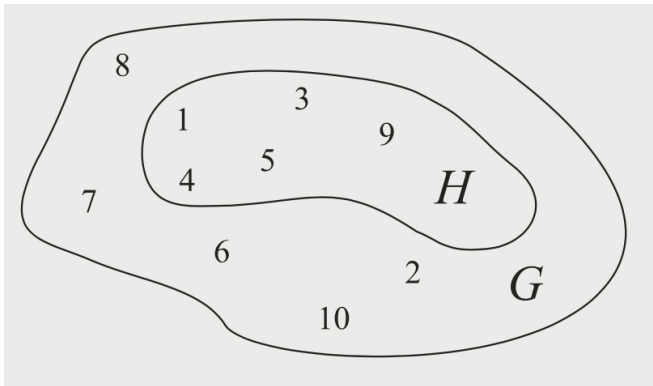
Example

Consider $G = \mathbb{Z}_{11}^*$ and $S = \{1, 2, 3\}$. Then S is not a subgroup because

- $2 \cdot 3 \bmod 11 = 6 \notin S$, violating the closure property.
- $3^{-1} \bmod 11 = 4 \notin S$, violating the invertibility property.

However, $\{1, 3, 4, 5, 9\}$ is a subgroup. You can verify it!

Subgroup of $\mathbb{Z}_{11}^*$



Definition (Order of an element)

The **order** $\text{ord}(a)$ of an element a of a group $(G, \circ)$ is the smallest positive integer k such that

$$a^k = 1 = \underbrace{a \circ a \circ \cdots \circ a}_{k \text{ times}},$$

where 1 is the identity element of G .

Example: Determining the Order in $\mathbb{Z}_{11}^*$

Example

We determine the order of $a = 3$ in the group $\mathbb{Z}_{11}^*$ by computing powers of a until we obtain the identity element 1:

$$a^1 = 3,$$

$$a^2 = a \cdot a = 3 \cdot 3 = 9,$$

$$a^3 = a^2 \cdot a = 9 \cdot 3 = 27 \equiv 5 \pmod{11},$$

$$a^4 = a^3 \cdot a = 5 \cdot 3 = 15 \equiv 4 \pmod{11},$$

$$a^5 = a^4 \cdot a = 4 \cdot 3 = 12 \equiv 1 \pmod{11}.$$

Therefore, $\text{ord}(3) = 5$.

Subgroup Generated by $g \in G$

Definition

For an element $g \in G$ with order n , we define

$$\langle g \rangle = \{g^0, g^1, \dots, g^{n-1}\}.$$

This is a subgroup of G (Why?); and its order is $\text{ord}(g) = n$.

Theorem (Lagrange's Theorem)

Let S be a subgroup of G . Then $|H|$ divides $|G|$.

Proposition

The order $\text{ord}(g)$ of g is always a divisor of $|G|$.

Example

If $G = \mathbb{Z}_{11}^*$ then

- $|G| = 10$
- $\text{ord}(2) = 10$ is a divisor of 10
- $\text{ord}(5) = 5$ is a divisor of 10

Subgroup Generated by an Element

Consider $G = \mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

i	0	1	2	3	4	5	6	7	8	9	10
$2^i \bmod 11$	1	2	4	8	5	10	9	7	3	6	1
$5^i \bmod 11$	1	5	3	4	9	1	5	3	4	9	1

Then,

$$\langle 2 \rangle = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\langle 5 \rangle = \{1, 3, 4, 5, 9\}.$$

Definition

An element $g \in G$ is a generator (or primitive element) if $\langle g \rangle = G$.

Proposition

g is a generator if and only if $\text{ord}(g) = |G|$.

Definition

G is a cyclic group if it has a generator.

Consider $G = \mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

i	0	1	2	3	4	5	6	7	8	9	10
$2^i \bmod 11$	1	2	4	8	5	10	9	7	3	6	1
$5^i \bmod 11$	1	5	3	4	9	1	5	3	4	9	1

Then,

$$\langle 2 \rangle = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\langle 5 \rangle = \{1, 3, 4, 5, 9\}.$$

- Is 2 a generator?
- Is 5 a generator?
- Is the group $\mathbb{Z}_{11}^*$ cyclic?

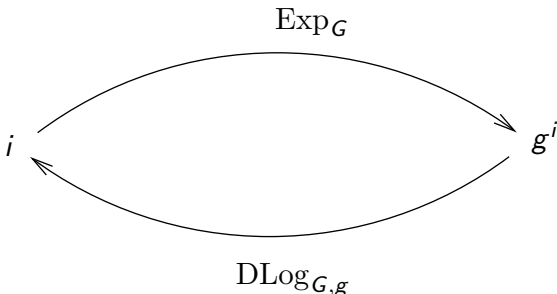
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Discrete Logarithm

If $G = \langle g \rangle$ is a cyclic group, then for every element $a \in G$, there is a unique exponent $i \in \{0, \dots, |G| - 1\}$ such that $g^i = a$. We call i the discrete logarithm of a to the base g and denote it by

$$\text{DLog}_g(a)$$

The discrete logarithm is the inverse function of the exponential function.



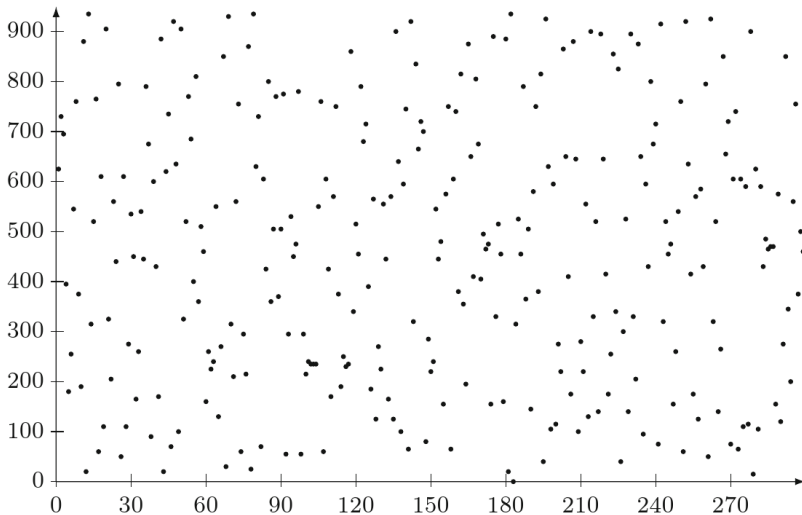
Consider $G = \mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. We know that 2 is a generator.

i	0	1	2	3	4	5	6	7	8	9	10
$2^i \bmod 11$	1	2	4	8	5	10	9	7	3	6	1

Then,

a	1	2	3	4	5	6	7	8	9	10
$\text{DLog}_2(a)$	0	1	8	2	4	9	7	3	6	5

Randomness of the powers $627^i \pmod{941}$

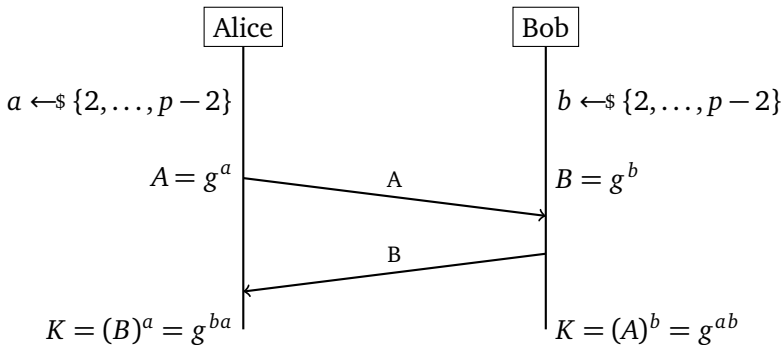


You are given a prime $p = 4969$ and the corresponding multiplicative group $\mathbb{Z}_{4969}^*$. You may use a computer for the computations.

- 1 Determine how many generators exist in $\mathbb{Z}_{4969}^*$.
- 2 What is the probability of a randomly chosen element $g \in \mathbb{Z}_{4969}^*$ being a generator?
- 3 Determine the smallest generator $g \in \mathbb{Z}_{4969}^*$ with $a > 1000$.

Recall: The Diffie-Hellman protocol

Public parameter: g, p



- Eavesdropper sees:

$$p, g, A = g^a \bmod p, \text{ and } B = g^b \bmod p$$

- Can she compute $g^{ab} \bmod p$?
- More generally, we define

$$\text{DH}_g(g^a, g^b) = g^{ab} \bmod p$$

- How hard is the DH function $\bmod p$?

How hard is the DH function $(\text{mod } p)$?

- Suppose prime p is n bits long.
- Best known algorithm (GNFS): run time $\exp(\text{ord}(\sqrt[3]{n}))$

Cipher key size	Modulus size	Elliptic Curve size
80 bits	1024 bits	160 bits
128 bits	3072 bits	256 bits
256 bits (AES)	15360 bits	512 bits

- As a result: slow transition away from $(\text{mod } p)$ to elliptic curves

Compute the following values in $\mathbb{Z}_{13}^*$:

- $\text{DH}_7(10, 5)$
- $\text{DH}_2(12, 9)$

Idea: Attack Diffie–Hellman

If Oscar could efficiently compute discrete logarithms in $\mathbb{Z}_p^*$, he could break Diffie–Hellman:

$$\begin{aligned}a &= \text{DLog}_g(A) \pmod{p}, \\ \text{DH}_g(A, B) &= B^a \pmod{p}.\end{aligned}$$

Thus

$$\text{DLog} \implies \text{DH}$$

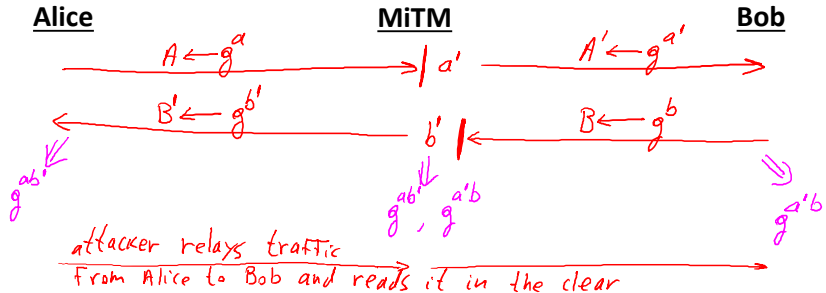
But discrete logs are believed to be hard.

Records for Computing Discrete Logarithms in $\mathbb{Z}_p^*$

Decimal digits	Bit length	Year
58	193	1991
65	216	1996
85	282	1998
100	332	1999
120	399	2001
135	448	2006
160	532	2007
180	596	2014
232	768	2016
240	795	2019

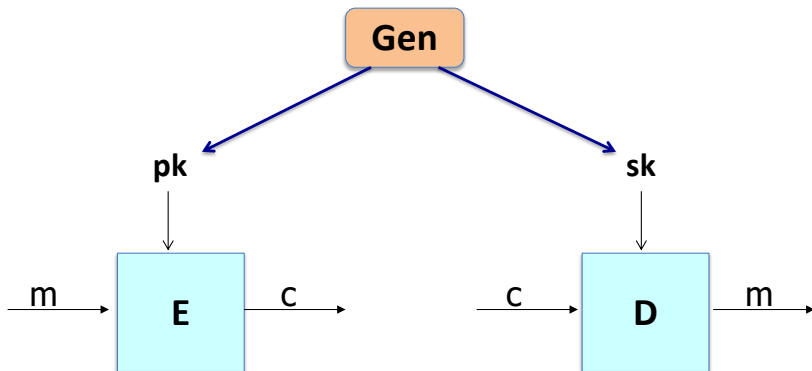
Some records for computing discrete logarithms in $\mathbb{Z}_p^$.*

Insecure against man-in-the-middle



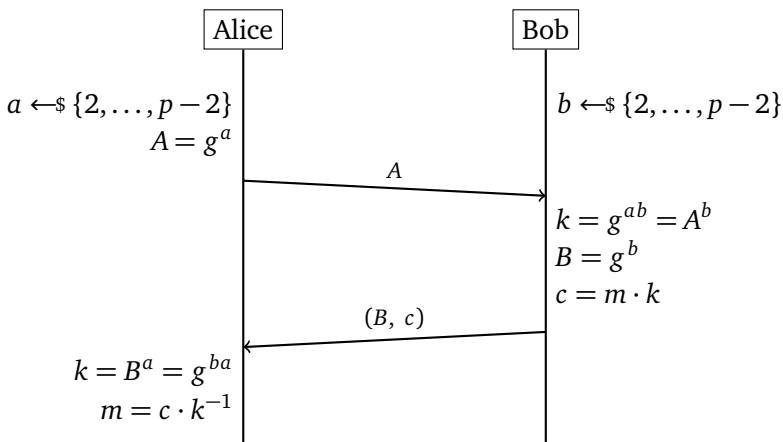
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Recap: public key encryption



ElGamal: converting to pub-key encryption

Public parameter: g, p



The ElGamal Scheme: Idea

Let Alice have public key g^a and secret key a .

If Bob wants to encrypt m for Alice, he

- Picks b and computes $k = g^{ab} = (g^a)^b$
- Sends ciphertext $c = (g^b, m \cdot k)$ to Alice.

Alice can recompute $k = g^{ab} = (g^b)^a$ because

- g^b is in the received ciphertext
- a is her secret key

and she can decrypt $m = c \cdot k^{-1}$.

Encrypt the following messages with the Elgamal scheme ($p = 467$ and $g = 2$):

- 1 secrete key $a = 105$ and $b = 213$ and message $m = 33$
- 2 secrete key $a = 105$ and $b = 123$ and message $m = 33$
- 3 secrete key $a = 300$ and $b = 45$ and message $m = 248$
- 4 secrete key $a = 300$ and $b = 47$ and message $m = 248$

ElGamal system (a modern view)

- G : finite cyclic group of order n
- (E_s, D_s) : symmetric encryption defined over (K, M, C) ;
- $H : G^2 \rightarrow K$: a hash function.

We construct a pub-key encryption system (G, E, D) .

ElGamal system (a modern view)

Key generation $G()$:

- choose random generator g in G and random a in $\mathbb{Z}_n$
- output $sk = a$, $pk = (g, h = g^a)$

Encryption $E(pk = (g, h), m)$:

$b \leftarrow \mathbb{Z}_n$, $u = g^b$, $v = h^b$, $k = H(u, v)$, $c \leftarrow E_s(k, m)$
return (u, c)

Decryption $D(sk = a, (u, c))$:

$v = u^a$, $k = H(u, v)$, $m = D_s(k, c)$
return m

Encryption & Decryption

$E(pk = (g, h), m) :$

$$b \leftarrow \mathbb{Z}_n, u = g^b, v = h^b$$

$D(sk = a, (u, c)) :$

$$v = u^a$$

Encryption: 2 exp.

(fixed basis)

- Can pre-compute $\{g^{(2^i)}, h^{(2^i)} \mid \text{for } i = 1, \dots, \log_2 n\}$
- 3x speed-up (or more)

Decryption: 1 exp.

(variable basis)

Computational Diffie-Hellman Assumption

- G : finite cyclic group of order n
- Computational Diffie-Hellman assumption holds in G if:

$$g, g^a, g^b \not\Rightarrow g^{ab}.$$

For all efficient algorithms A :

$$\Pr[A(g, g^a, g^b) = g^{ab}] < \text{negligible}$$

where $g \leftarrow \{ \text{generators of } G \}$, $a, b \leftarrow \mathbb{Z}_n$.



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Thank you!

